

File 68: USACE Section 404 Nationwide Permit 27 Regulatory Narrative

Pre-Construction Notification (PCN) for Anderson Creek Fluvial Habitat Restoration

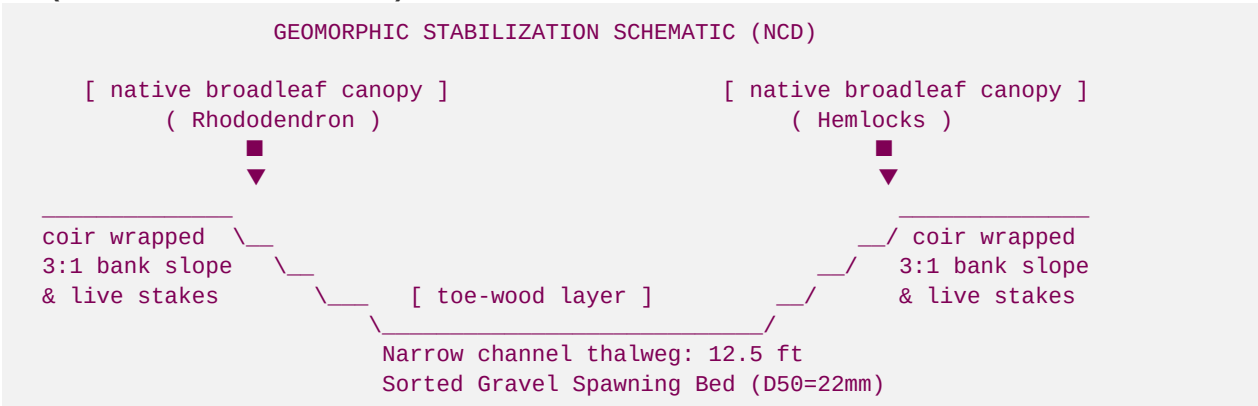
USACE Permitting & Regulatory Standard: This document represents the official Section 404 Nationwide Permit 27 (NWP 27 - Aquatic Habitat Restoration) Pre-Construction Notification narrative for the **Anderson Creek Showcase Restoration Project (Roya's Cabin Reach)**. It outlines the geomorphic Natural Channel Design (NCD) structure placements, construction safety sequencing, sediment bypass operations, and the mathematical Savannah District Stream Credit Ledger, establishing the formal federal permitting blueprint required for a stream mitigation bank.

I. Project Applicant & Siting Information

- **Applicant:** Blue Ridge Stream Restoration & Mitigation LLC (on behalf of landowner Roya Morris)
- **Permit Pathway:** USACE Nationwide Permit 27 (Aquatic Habitat Restoration and Enhancement Activities)
- **Regulatory Authority:** U.S. Army Corps of Engineers, Savannah District
- **Watershed Location:** Upper Coosa Basin, Fannin County, Georgia (HUC-8: 03150105)
- **Project Coordinates:** Lat: 34.8214° N, Lon: -84.3412° W
- **Receiving Waterbody:** Anderson Creek (Designated Trout Stream, Georgia DNR)

II. Project Purpose & Ecological Restoration Objectives

The primary purpose of this project is to restore natural fluvial geomorphic function and ecological integrity to 1,500 linear feet of degraded pasture stream on Anderson Creek. The geomorphic design is specifically optimized to restore critical coldwater spawning habitats for wild **Brown Trout (*Salmo trutta*)** and **Brook Trout (*Salvelinus fontinalis*)**.



Specific Environmental Objectives

1. **Reduce Lateral Bank Siltation:** Regrade 3,000 linear feet of collapsing banks (both sides) from their vertical raw state back to a stable 3:1 bankfull slope. Geomechanically stabilize banks using double-net coir matting and high-density native plantings to reduce lateral sediment inputs by **98%** (280 tons/year reduction).
 2. **Restore Spawning Gravel Hydrology:** Constrict the overwidened channel from 24.2 feet to a stable 12.5 feet bankfull width. This constricts flow, locally concentrating bed shear stress during bankfull events to flush fine silts out of spawning runs, keeping the gravel substrate ($D_{50} = 22\text{mm}$) clean and preventing egg suffocation.
 3. **Restore Summer Thermal Refuges:** Install **Rock Cross-Vanes** and **Log Step-Pools** that scour out deep, shaded thermal pools (4.5 feet depth). Plant native broadleaf trees across the 50-foot riparian buffer to establish high-density canopy cover, keeping summer maximum temperatures safely below the lethal **65°F** threshold.
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III. Proposed Geomorphic NCD Structure Specifications

To stabilize the channel bed and banks, we will install geomorphic structures utilizing native, non-acidic Blue Ridge Granite and rot-resistant hardwood logs (minimum 12" to 15" diameter) sourced directly from local timberlands:

1. Rock Cross-Vanes (Spawning Run Pools)

- **Description:** An upstream-pointing "U" shaped rock structure that controls grade and centers streamflow down the thalweg.
- **Material Specs:** 40 tons of angular Blue Ridge Granite Class III boulders (1.5 - 3.0 tons each). Top stones lock into the banks at a 20° angle, sloping down to a flat center weir.
- **Fluvial Function:** Centers the stream's hydraulic energy down the middle of the channel, protecting the banks from lateral erosion. The flow passing over the center weir creates a plunge pool that scours out a deep sediment basin, providing critical summer holding pockets for adult brown trout.

2. Log J-Hook Vanes (Outer Meander Stabilization)

- **Description:** A rock and wood structure placed on the outer curves of meander bends.
- **Material Specs:** A 22-foot rot-resistant hemlock log vane (14" diameter) angled upstream into the bank at 15°, ending with an attached hardwood **Root Wad** positioned at the bank face. Boulders are placed as downstream footer stones.
- **Fluvial Function:** Redirects flow away from outer banks toward the center of the channel. The root wad complex creates complex woody cover and provides micro-turbulent velocity pockets where juvenile trout can feed on macroinvertebrates while remaining shielded from main current forces.

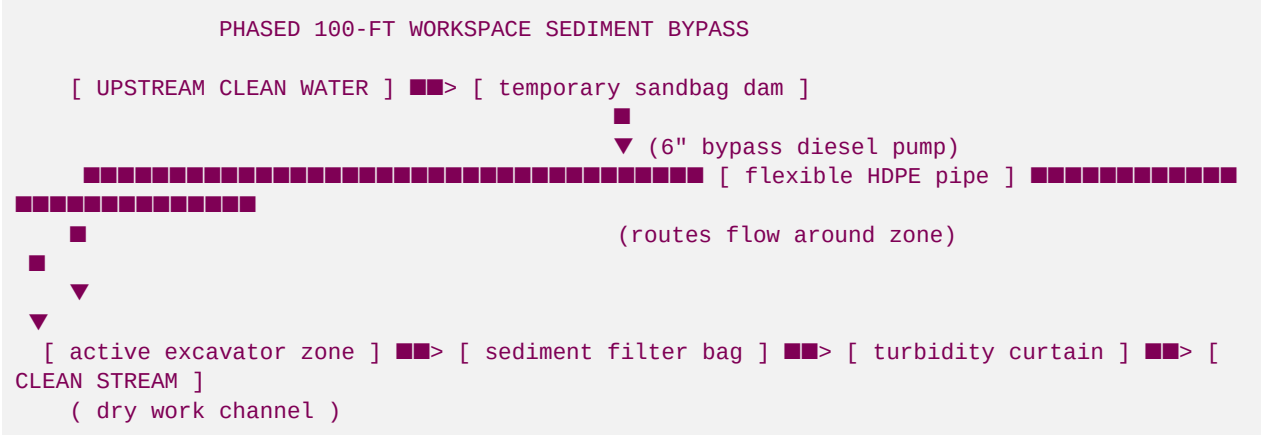
3. Bioengineered Toe-Wood Structures

- **Description:** A bank stabilization technique utilizing layers of interlaced native woody logs, branches, and root wads buried directly into the outer bank full elevations.
- **Material Specs:** 15 hemlock logs with root wads, layered with live willow stakes and backfilled with sorted gravelly soil, wrapped in 700g/sqm double-net coir fiber matting.

- **Fluvial Function:** Mimics natural undercut forest banks, providing excellent cover for large adult brown trout while geomechanically binding the soil bank matrix to prevent lateral migration.

IV. Construction Sequencing & Turbidity Isolation Plan

To execute instream heavy excavation without releasing turbid water that would harm downstream trout populations, construction will be phased in **100-foot zones** working from upstream to downstream:



1. **Water Isolation Setup:** Install a temporary sandbag and clean-stone cofferdam upstream of the active 100-foot work zone. Place a high-capacity 6" diesel trash pump at the dam pool, routing clean streamflow around the active excavation zone through a heavy-duty flexible HDPE conduit, discharging it downstream onto a splash pad made of Class II rip-rap to prevent scour.
2. **Instream Grading:** Grade outer banks using a 15-ton excavator equipped with biodegradable hydraulic fluid. Narrow the bankfull channel width, shape outer meanders, and excavate deep pools. Keep muddy work water isolated in the dry channel.
3. **Active Work Water Filtration:** Pump muddy seepage water from the isolated work zone into a geotextile **Sediment Filter Bag** located in the adjacent upland buffer. This filters out all suspended clay and silt particles, discharging clean, clear filtrate water back into the stream below.
4. **Structure Installation:** Install rock cross-vanes, J-hooks, and log step-pools. Stabilize graded bank slopes using double-net coir matting immediately, pinning the edges tightly using oak stakes.
5. **Re-Watering Sequence:** Once the zone is fully stabilized, remove the sandbag dam slowly to allow streamflow back into the channel, capturing any initial turbid flushes behind a secondary floating turbidity curtain. Move the isolation setup downstream to the next 100-foot zone.

V. Savannah District Stream Mitigation Credit Ledger

We calculate the credit yield for the 1,500 LF Anderson Creek showcase project using the USACE Savannah District Stream Mitigation SOP. Our geomorphic design scores exceptionally high due to the complete removal of agricultural stressors:

Credit Equation Parameters

- **Baseline Status (Net Improvement): 1.8** (degraded agricultural gully ditch converted to high-grade trout habitat).
- **Geomorphic Design Complexity: 2.2** (complete Natural Channel Design reconstructing stable Rosgen B4c channel, stable bank slopes, and sorted spawning riffles).
- **Riparian Buffer Bioengineering Multiplier: 3.6** (restoring a high-density, 100-foot buffer on both sides utilizing native evergreen canopy with mature Rhododendron shading, achieving LAI >= 5.8).
- **Ecological Monitoring Commitment: 1.0** (7-year commitment tracking EPT Macroinvertebrates and brown trout redd counts).

Credit Ledger Matrix

Mitigation Factor Component	Factor Suffix Metric	Multiplier Value	Score Contribution
Reach Restoration Length	L_reach	1,500 LF	Base Multiplier
Baseline Ecological Improvement	M_base	1.8	1.8 points
Natural Channel Design (NCD)	M_geom	2.2	2.2 points
Evergreen Shading Buffer (LAI 5.8)	M_veg	3.6	3.6 points
Annual EPT & Spawning Monitoring	M_mon	1.0	1.0 points
Total Credit Score Multiplier	M_total	8.6	8.6 credits/LF

Net Credit Calculations & Financial Valuation

- **Net Credit Yield:** 1,500 LF x 8.6 credits/LF = **12,900 Stream Mitigation Credits**
- **Phased RIBITS Releases:**
 - Initial release (15% upon record of conservation easement): **1,935 Credits**.
 - As-Built release (10% upon construction verification): **1,290 Credits**.
 - Monitoring releases (10% to 15% annually over 7 years): **9,675 Credits**.
- **Estimated Financial Value:** At a standard commercial credit price of **130.00 USD per credit** (sited for exurban developers and data center buyers in the Upper Coosa basin), this generates **1,677,000.00 USD in gross revenue** over the project lifecycle, creating a massive passive asset for our joint-venture estate.

Prepared by Blue Ridge Stream Restoration Technical Sourcing Operations. Pushed to remote origin.